

MIAE 221: Materials Science for Engineers

Part 4: Crystallographic Densities, XRD, Anisotropy & Bragg's Law Master Guide

Executive Overview & Core Concepts

In solid-state materials engineering, macroscopic mechanical, physical, and optical behavior stems directly from how atoms populate specific spatial lines and crystallographic planes. This master guide covers the rigorous foundations of:

1. **Linear and Planar Atomic Densities:** Quantitative derivations of atom packing along direction vectors and planes.
 2. **Directional Anisotropy vs. Polycrystalline Isotropy:** Why elasticity varies with direction in single crystals (e.g., turbine blades) while bulk polycrystalline metals behave isotropically.
 3. **Interplanar Spacing & X-Ray Diffraction (XRD):** The wave-optics principles of constructive interference and the mathematical derivation of Bragg's Law.
 4. **Powder Diffractometry & Crystal Identification:** How diffraction spectra and selection rules reveal unknown lattice parameters and crystal structures.
 5. **Polymorphism & Allotropy:** Temperature/pressure-induced structural phase shifts in carbon and iron.
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1. Linear Atomic Density (LD)

Definition & Physical Principle

Linear Density (LD) is the number of atom diameters intercepted per unit length along a specific crystallographic direction vector:

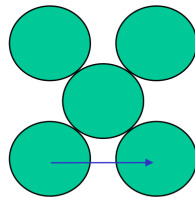
$$LD = \frac{\text{Number of atomic diameters centered on the direction vector}}{\text{Length of the direction vector}} = \frac{n}{L}$$

- **Counting Rule:** For an atom to be counted toward linear density, its **center must lie directly on the direction vector**.
 - End atoms whose centers terminate a line segment inside a unit cell contribute $\frac{1}{2}$ atom each.
 - Atoms whose centers lie strictly along the interior of the line segment contribute 1 full atom each.
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Linear Density

- Calculate the linear density of the [100] direction for the FCC crystal



$$L_D = n/L_L \quad \text{linear density}$$

$$n = 1 \quad \text{atoms}$$

$$L_L = a \quad \text{line length}$$

For FCC $a = \dots\dots\dots$

$$L_D = \dots\dots\dots$$

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Figure 4.1: Linear density calculation along the [100] direction in a Face-Centered Cubic (FCC) unit cell (Dr. Medraj MIAE 221 Lecture 6). Atoms are centered only at the two cube corners along the edge length a .

- Direction Vector Length:** The direction [100] lies along the edge of the cubic unit cell.

$$L = a$$

In FCC, the lattice parameter relates to atomic radius R by $a\sqrt{2} = 4R \implies a = 2\sqrt{2}R$.

$$L = 2\sqrt{2}R$$

- Atoms Centered on the Vector:** The vector passes through two corner atoms. The face-center atoms do not lie on the edge.

$$n = 2 \times \frac{1}{2} = 1 \text{ atom}$$

- Linear Density Formula:**

$$LD_{[100]} = \frac{1}{a} = \frac{1}{2\sqrt{2}R}$$

- Numerical Example (FCC Aluminum):** Given atomic radius $R = 0.143 \text{ nm} = 0.143 \times 10^{-7} \text{ cm}$:

$$a = 2\sqrt{2}(0.143 \text{ nm}) = 0.4045 \text{ nm}$$

$$LD_{[100]} = \frac{1}{0.4045 \text{ nm}} = 2.47 \text{ nm}^{-1} = 2.47 \times 10^7 \text{ cm}^{-1}$$

Close-Packed Direction: FCC [110]

Along the $[110]$ face diagonal:

- Atoms touch continuously: $L = 4R$.
- Atom count: $2 \text{ corners} \times \frac{1}{2} + 1 \text{ face center} = 2 \text{ atoms}$.
- Linear Density:

$$LD_{[110]} = \frac{2}{4R} = \frac{1}{2R}$$

- For Aluminum: $LD_{[110]} = \frac{1}{2(0.143 \text{ nm})} = \mathbf{3.50 \text{ nm}^{-1}}$.
 - **Physical Insight:** $LD_{[110]} > LD_{[100]}$ ($3.50 \text{ nm}^{-1} > 2.47 \text{ nm}^{-1}$). The $[110]$ direction is the **close-packed direction** in FCC; this is why plastic deformation (dislocation glide) occurs preferentially along $\langle 110 \rangle$.
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2. Planar Atomic Density (PD)

Definition & Physical Principle

Planar Density (PD) is the number of atoms centered on a crystallographic plane per unit area of that plane:

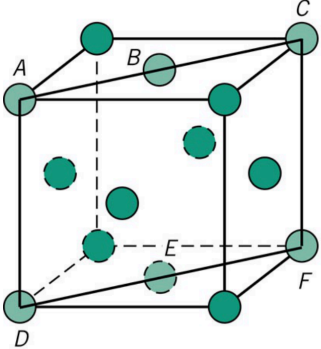
$$PD = \frac{\text{Number of atoms centered on the plane}}{\text{Area of the plane}} = \frac{n}{A}$$

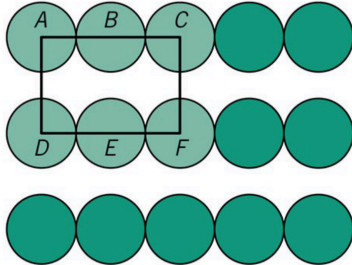
- **Centering Law:** Only atoms whose centers lie strictly within the geometric boundary of the plane are counted.
 - Corner atoms of a planar polygon contribute fractional areas based on their interior plane angles.
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Planar Density

- Calculate the planar density of the (110) plane for the FCC crystal





- Compute planar area
- Compute number of atoms

For an atom to be counted, it has to be centered on that plane.

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Figure 4.2: Planar density geometry for the (110) plane in an FCC crystal (Dr. Medraj Lecture 6). The plane intersects the unit cell as a rectangle of dimensions a by $a\sqrt{2}$.

1. **Planar Geometry & Area:** The (110) plane cuts diagonally across the FCC unit cell, forming a rectangle:

- Vertical height = a
- Horizontal width = Face diagonal = $a\sqrt{2}$

$$\text{Area } A = a \times a\sqrt{2} = a^2\sqrt{2}$$

Substituting $a = 2\sqrt{2}R$:

$$A = (2\sqrt{2}R)^2\sqrt{2} = 8R^2\sqrt{2} = 8\sqrt{2}R^2$$

2. **Number of Centered Atoms (n):**

- 4 corner atoms: each interior corner angle of the rectangle is 90° ($\frac{90^\circ}{360^\circ} = \frac{1}{4}$ of a circle).

$$4 \times \frac{1}{4} = 1 \text{ atom}$$

- 2 face-centered atoms: located on the top and bottom faces, each halved by the boundary edge of the rectangle.

$$2 \times \frac{1}{2} = 1 \text{ atom}$$

- Total centered atoms:

$$n = 1 + 1 = 2 \text{ atoms}$$

3. Planar Density Formula:

$$PD_{(110)} = \frac{n}{A} = \frac{2}{8\sqrt{2}R^2} = \frac{1}{4\sqrt{2}R^2}$$

4. Numerical Example (FCC Aluminum): With $R = 0.143 \text{ nm}$:

$$PD_{(110)} = \frac{1}{4\sqrt{2}(0.143 \text{ nm})^2} = \frac{1}{4(1.4142)(0.02045 \text{ nm}^2)} = \mathbf{8.64 \text{ atoms/nm}^2}$$

Engineering Significance of LD and PD

1. Slip Systems & Plastic Deformation:

- Dislocation motion requires overcoming lattice friction (Peierls-Nabarro stress $\tau_{PN} \propto \exp(-2\pi d/b)$).
- Planes with the **highest planar density** have the widest interplanar spacing (d), minimizing lattice resistance.
- Consequently, plastic slip occurs along **directions of highest linear density** within **planes of highest planar density**:
 - **FCC Slip System:** $111\langle 110 \rangle$ (12 slip systems $\implies$ high ductility).
 - **BCC Slip System:** $110\langle 111 \rangle$ (48 active slip systems at elevated temperatures).

2. Speed of Sound: Sound waves propagate via phonon atomic collisions; acoustic velocities are highest along densely packed directions.

3. Surface Energy & Catalysis: Low planar density planes expose more unsatisfied ("dangling") chemical bonds, resulting in higher surface energy and heightened catalytic reactivity.

3. Directional Anisotropy vs. Polycrystalline Isotropy

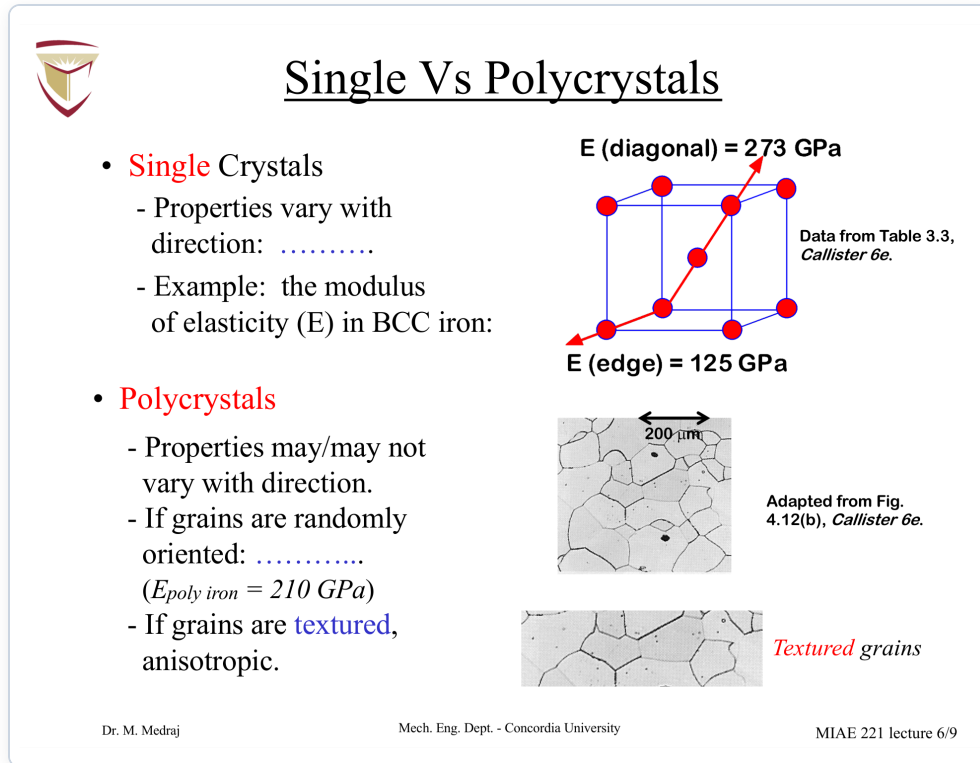


Figure 4.3: Directional dependence (anisotropy) of the Elastic Modulus E in single-crystal BCC Iron (Dr. Medraj Lecture 6). Left: 3D spatial stiffness surface showing dramatic variation between $[111]$ and $[100]$. Right: Polycrystalline aggregate exhibiting microscopic grain boundaries and macroscopic isotropy.

Single Crystals & Anisotropy

In a **single crystal**, the periodic arrangement of atoms is continuous across the entire volume without interruption:

- Physical properties (stiffness, electrical conductivity, magnetic susceptibility, refractive index) depend on the crystallographic direction along which they are measured. This directionality is called **anisotropy**.
- **BCC Iron (Fe) Case Study:**
 - $[111]$ direction (body diagonal, close-packed): $E_{[111]} = 272.7 \text{ GPa}$ (stiffest direction).
 - $[110]$ direction (face diagonal): $E_{[110]} = 210.5 \text{ GPa}$.
 - $[100]$ direction (cube edge, most open): $E_{[100]} = 125.0 \text{ GPa}$ (most compliant direction).
 - Ratio: $\frac{E_{[111]}}{E_{[100]}} = 2.18$! A single crystal of iron is more than twice as stiff along its body diagonal as along its cube edge.

High-Performance Single Crystal Applications

- **Turbine Blades:** Modern aerospace jet engines operate at temperatures exceeding 1100°C . High-temperature creep occurs primarily via atomic diffusion along grain boundaries. By casting turbine blades as **single crystals** (e.g., PWA 1480, CMSX-4 nickel-base superalloys) oriented along the stiff $[001]$ growth direction, grain boundaries are eliminated, preventing creep failure.
- **Diamond Abrasives:** Industrial cutting tools exploit diamond's maximum hardness along specific crystallographic planes.

Polycrystalline Aggregates & Isotropy

- Most engineering metals are **polycrystals** consisting of millions of microscopic grains separated by grain boundaries.
- If the grains are randomly oriented (equiaxed structure), directional variations average out over macroscopic dimensions, yielding **macroscopic isotropy**.
- When metals undergo heavy plastic deformation (e.g., cold rolling of sheet steel), grains develop a preferred crystallographic orientation (**texture**), reintroducing macroscopic anisotropy.

4. Interplanar Spacing (d_{hkl})

In any crystal lattice, parallel atomic planes designated by Miller indices (hkl) are separated by a constant perpendicular distance called the **interplanar spacing** (d_{hkl}).

For **cubic crystal systems** ($a = b = c$):

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Key Geometric Relationships

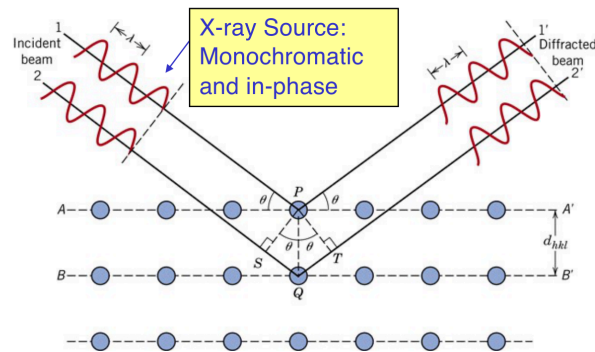
- Higher Miller index planes have **smaller** interplanar spacings.
- For a cubic unit cell with lattice parameter a :
 - $d_{100} = \frac{a}{\sqrt{1^2+0^2+0^2}} = a$
 - $d_{110} = \frac{a}{\sqrt{1^2+1^2+0^2}} = \frac{a}{\sqrt{2}} \approx 0.707a$
 - $d_{111} = \frac{a}{\sqrt{1^2+1^2+1^2}} = \frac{a}{\sqrt{3}} \approx 0.577a$
 - $d_{200} = \frac{a}{\sqrt{2^2+0^2+0^2}} = \frac{a}{2} = 0.500a$

5. X-Ray Diffraction (XRD) & Bragg's Law

X-ray diffraction is the definitive experimental technique used by materials scientists and engineers to identify unknown crystal structures, quantify lattice parameters, and measure residual stresses.



Bragg's Law



The law: For constructive interference, the additional path length $SQ + QT$ must be an integer number of wavelength (λ).

$$n\lambda = SQ + QT = d_{hkl}\sin\theta + d_{hkl}\sin\theta = 2 d_{hkl}\sin\theta$$

$n = 1, 2, 3, \dots$ order of reflection

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Figure 4.4: Derivation of Bragg's Law for X-ray diffraction from parallel crystallographic planes (Dr. Medraj Lecture 6). Constructive interference requires path difference $SQ + QT = 2d_{hkl}\sin\theta = n\lambda$.

Physical Mechanism of Diffraction

- 1. Electromagnetic Wave Character:** X-rays are high-energy electromagnetic radiation with wavelengths ($\lambda \approx 0.05 - 0.2 \text{ nm}$ or $0.5 - 2 \text{ \AA}$) on the same spatial scale as interatomic crystal spacings ($d \approx 0.1 - 0.3 \text{ nm}$).
- 2. Elastic Scattering:** When an incident X-ray beam strikes an atom, its electrons scatter the wave elastically without changing its wavelength.
- 3. Constructive vs. Destructive Interference:**
 - Scattered rays from adjacent atomic planes travel different path lengths.
 - If the scattered waves emerge **out of phase**, destructive interference cancels their amplitude, yielding zero detected intensity.
 - If the scattered waves emerge **in phase**, constructive interference produces a sharp, intense diffracted beam.

Mathematical Derivation of Bragg's Law

Consider two parallel planes of atoms with interplanar spacing d_{hkl} struck by a monochromatic, in-phase X-ray beam at grazing angle θ :

- Ray 1 reflects from the top plane at point P .
- Ray 2 reflects from the second plane at point Q , traveling an additional distance equal to $SQ + QT$.

3. From right triangles $\triangle SPQ$ and $\triangle TPQ$:

$$\sin \theta = \frac{SQ}{d_{hkl}} \implies SQ = d_{hkl} \sin \theta$$

$$\sin \theta = \frac{QT}{d_{hkl}} \implies QT = d_{hkl} \sin \theta$$

4. Total path difference (Δ):

$$\Delta = SQ + QT = 2d_{hkl} \sin \theta$$

5. For the two waves to reinforce constructively, the extra distance must equal an integer number (n) of wavelengths:

$$\Delta = n\lambda$$

$$n\lambda = 2d_{hkl} \sin \theta$$

Where:

- $n = 1, 2, 3 \dots$ is the **order of reflection** (typically $n = 1$).
- λ = Wavelength of the incident X-ray beam (commonly $Cu-K_\alpha$: $\lambda = 0.15418 \text{ nm}$ or $Fe-K_\alpha$: $\lambda = 0.1790 \text{ nm}$).
- d_{hkl} = Interplanar spacing of the reflecting planes (hkl).
- θ = **Bragg angle** (half of the detector angle 2θ).

6. Powder Diffraction & Spectral Analysis

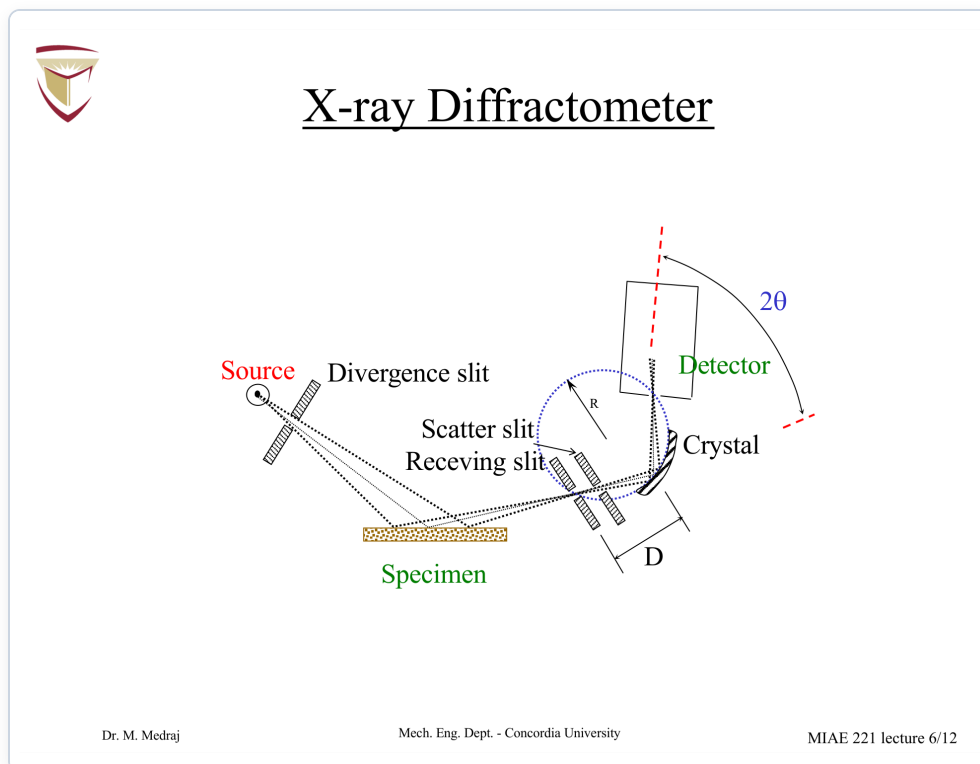


Figure 4.5: Goniometer geometry of an automated powder X-ray diffractometer (Dr. Medraj Lecture 6). The X-ray source, flat specimen, and scintillation detector rotate through angle 2θ .

The Powder Method

- A finely ground powder contains millions of microscopic, randomly oriented crystallites.
- By rotating the X-ray detector continuously across an angular sweep, every possible (hkl) planar family will eventually satisfy Bragg's condition ($2d \sin \theta = \lambda$) at its specific angle 2θ .

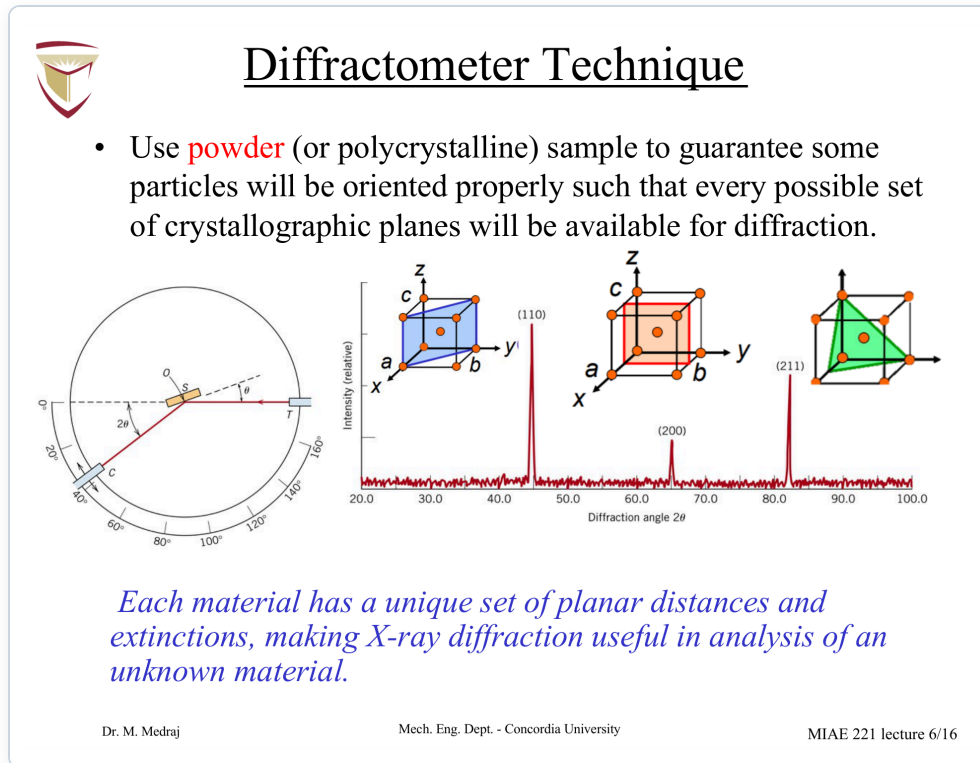


Figure 4.6: Representative powder XRD diffractogram plotting diffracted intensity vs. diffraction angle 2θ (Dr. Medraj Lecture 6). Each sharp peak corresponds to constructive diffraction from a specific crystallographic plane.

Systematic Reflection Rules (Extinction Conditions)

Due to destructive interference caused by atoms residing at center or face locations within a unit cell, certain (hkl) reflections have exactly zero intensity:

Crystal Structure	Allowed Reflections (Peaks Appear)	Forbidden Reflections (Extinct / Zero Intensity)
BCC (Body-Centered Cubic)	$h + k + l = \text{Even}$ (110), (200), (211), (220), (310), (222) ...	$h + k + l = \text{Odd}$ (100), (111), (210), (300) ...
FCC (Face-Centered Cubic)	h, k, l are all odd OR all even (Unmixed) (111), (200), (220), (311), (222), (400) ...	h, k, l are mixed (even and odd) (100), (110), (210), (211) ...

[!IMPORTANT] Notice that the **first diffraction peak** for BCC is always (110), whereas the first diffraction peak for FCC is always (111). This enables instant identification of unknown crystal lattices from experimental XRD patterns!

7. Step-by-Step Quantitative Exam Problem

(Directly adapted from Dr. Medraj MIAE 221 Lecture 6, Slide 17)

Problem Statement

For Body-Centered Cubic (BCC) Iron (Fe):

1. Compute the **interplanar spacing** d_{220} of the (220) planes.
2. Determine the **diffraction angle** (2θ) at which the (220) reflection occurs.

Given Data:

- Lattice parameter of BCC Iron: $a = 0.2866 \text{ nm}$ ($2.866 \times 10^{-10} \text{ m}$)
- X-ray radiation wavelength: $\lambda = 0.1790 \text{ nm}$
- Reflection order: $n = 1$

Step-by-Step Solution

Part (a): Compute Interplanar Spacing d_{220}

Using the cubic spacing formula for $(hkl) = (220)$:

$$\begin{aligned}d_{hkl} &= \frac{a}{\sqrt{h^2 + k^2 + l^2}} \\d_{220} &= \frac{0.2866 \text{ nm}}{\sqrt{2^2 + 2^2 + 0^2}} = \frac{0.2866 \text{ nm}}{\sqrt{4 + 4 + 0}} = \frac{0.2866 \text{ nm}}{\sqrt{8}} \\ \sqrt{8} &= 2\sqrt{2} \approx 2.8284 \\d_{220} &= \frac{0.2866}{2.8284} = \mathbf{0.1013 \text{ nm}} = \mathbf{1.013 \text{ \AA}}\end{aligned}$$

Part (b): Compute the Diffraction Angle 2θ

Using Bragg's Law:

$$\lambda = 2d_{220} \sin \theta$$

Solve for $\sin \theta$:

$$\sin \theta = \frac{\lambda}{2d_{220}} = \frac{0.1790 \text{ nm}}{2(0.1013 \text{ nm})} = \frac{0.1790}{0.2026} \approx 0.8835$$

Compute θ :

$$\theta = \arcsin(0.8835) = 62.07^\circ$$

The diffractometer instrument records the **diffraction angle** 2θ :

$$2\theta = 2 \times 62.07^\circ = \mathbf{124.14^\circ}$$

8. Polymorphism and Allotropy



Polymorphism and Allotropy

- Some materials may have more than one **crystal structure** depending on **temperature** and **pressure** - called *POLYMORPHISM*
- Carbon
 - graphite
 - diamond
- Iron BCC and FCC

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Figure 4.7: Polymorphic structural allotropy (Dr. Medraj Lecture 6). Left: Carbon allotropes—Diamond (3D covalent network) vs. Graphite (2D hexagonal sheets). Right: Temperature-induced allotropic phase transformations in Iron.

Core Terminology

- **Polymorphism**: The phenomenon wherein a solid material can exist in more than one crystal structure depending on ambient temperature and pressure.
- **Allotropy**: Polymorphism occurring specifically in **elemental solids** (pure elements).

Case Study 1: Carbon Allotropes

1. Graphite:

- Atoms form 2D planar hexagonal arrays (sp^2 hybridized covalent bonding).

- Parallel sheets are held together by exceptionally weak van der Waals secondary bonds.

- *Properties*

: Soft, slippery (superb solid lubricant), highly electrically conductive within basal planes.

2. **Diamond:**

- Formed at high temperatures and extreme pressures ($> 1500^{\circ}\text{C}$, $> 5\text{ GPa}$).
- Every carbon atom forms 4 rigid tetrahedral sp^3 covalent bonds (109.5°).

- *Properties*

: Hardest known natural material (10 on Mohs scale), high thermal conductivity ($2000\text{ W/m}\cdot\text{K}$), wide-bandgap electrical insulator.

Case Study 2: The Allotropy of Iron (Fe)

Iron exhibits three distinct crystalline allotropes between room temperature and its melting point (1538°C):

Phase	Temperature Range	Crystal Structure	Coordination Number (CN)	Atomic Packing Factor (APF)	Engineering Significance
α -Ferrite	Up to 912°C	BCC	8	0.68	Ferromagnetic, low carbon solubility ($<0.022\text{ wt}\%$)
γ -Austenite	$912^{\circ}\text{C} - 1394^{\circ}\text{C}$	FCC	12	0.74 (Denser!)	Nonmagnetic, high carbon solubility (up to $2.14\text{ wt}\%$)
δ -Ferrite	$1394^{\circ}\text{C} - 1538^{\circ}\text{C}$	BCC	8	0.68	High-temperature stable BCC phase prior to melting

[!WARNING] **Volume Change During Heating ($\alpha \rightarrow \gamma$):** When iron is heated through 912°C , it undergoes a phase transformation from BCC to FCC. Because FCC packs atoms more densely ($\text{APF} = 0.74$) than BCC ($\text{APF} = 0.68$), iron **contracts in volume** upon heating through 912°C ! This phase transition is the fundamental basis for all steel heat treatment (quenching, tempering, martensitic transformation).

9. Master Synthesis & Exam Traps Matrix

Exam Trap / Critical Concept	Common Student Error	Correct Physical Principle
Diffraction Angle (θ vs. 2θ)	Reporting θ as the final answer	The diffractometer records the detector angle 2θ . Always multiply θ by 2!
Centering Atoms in Planar Density	Counting corner atoms on adjacent unit cells	Only atoms whose centers lie within the plane boundary contribute.
BCC Systematic Extinctions	Expecting (100) or (111) peaks in BCC iron	$h + k + l$ must be even! (110) is the first allowed reflection; (100) produces complete destructive interference.
FCC Systematic Extinctions	Expecting (110) peaks in FCC aluminum	h, k, l must be unmixed! (111) is the first peak; (110) is mixed (even+odd) and extinct.
Single Crystal vs. Polycrystal	Assuming all metals are anisotropic	Randomly oriented polycrystals exhibit macroscopic isotropy because directional variations cancel out.